

Hritvik Arora

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EDUCATION

Indian Institute of Science Education and Research (IISER) Kolkata 2024 – Present
BS-MS Dual Degree Programme CGPA: 7.91/10

City Montessori School, Lucknow 2021 – 2023
ISC (Class XII) 94%

City Montessori School, Lucknow 2019 – 2021
ICSE (Class X) 96.4%

RESEARCH INTERESTS

- Computational Geoscience • Computational Seismology • Geophysics • Scientific Computing
- Numerical Modelling • Seismic Imaging • Scientific Image Processing • Digital Petrography
- Quantitative Microstructural Analysis • Structural Geology

RESEARCH EXPERIENCE

Summer Research Intern May 2026 – June 2026
Indian Institute of Technology Roorkee Supervisor: Prof. Sudheer Kumar Tiwari

- Assisted in computational analysis of petrographic datasets for structural geology applications.
- Performed MATLAB-based scientific programming and data analysis for geological applications.
- Prepared technical documentation and research reports related to ongoing research activities.

PROJECTS

GrainSegmenter – MATLAB based Software for Petrographic Image Analysis

- Developed a MATLAB-based software tool for semi-automated grain boundary segmentation of petrographic thin section images.
- Designed an interactive graphical interface supporting manual boundary refinement through draw/erase tools, adjustable segmentation tolerance, undo/redo operations, zoom controls, and project persistence.
- Reduced manual effort in petrographic image analysis by integrating automated segmentation with efficient user guided boundary refinement.

MYflow Software Validation and Enhancement (MATLAB)

- Performed algorithmic validation of the MYflow composite-mode workflow to investigate grain count inconsistencies in petrographic phase-map analysis.
- Improved the grain identification pipeline by redesigning boundary detection from the original RGB image and enhancing phase assignment, connected-component analysis, and overall workflow robustness.
- Developed multiple reliability improvements including input validation, duplicate-phase protection, zero-label filtering, histogram edge-case handling, and enhanced error management.

- Authored a technical validation report documenting the investigation, code modifications, and experimental evaluation. The work was acknowledged and appreciated by Dr. Leonardo Casini, developer of MYflow.

TECHNICAL SKILLS

Programming: MATLAB, Python, Java, LaTeX

Scientific Computing: Image Processing, Digital Image Analysis, Numerical Modelling, Data Analysis

Software: MATLAB, QGIS, Git, GitHub, Adobe Illustrator, Microsoft Excel, SnapGene

POSITIONS OF RESPONSIBILITY

Fest Coordinator — Inquivesta XIII, IISER Kolkata

- Leading the planning, coordination, and execution of the institute's annual science fest, overseeing 9+ organizing teams and coordinating 16+ events with 400+ expected participants.

Treasurer — Cricket Club, IISER Kolkata

- Managed club finances and coordinated tournaments.

Editor — PREP4IISER

- Formatted previous-year examination papers and solutions using LaTeX.
- Maintained publication-quality formatting across educational material.

Sports and Extra-Curricular

- An active institute level Cricketer.
- Represented IISER Kolkata as a Wicket Keeper Batsman in Inter Institute Sports Meet 2025.

RELEVANT COURSEWORK

Structural Geology, Geophysics, Data Structure and Algorithm, Petrology, Mineralogy, Numerical Methods, Advanced Programming in Python, Remote Sensing, Earth System Science.